

Daily Logic Drill #3

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Direct Proof · Proof by Contrapositive.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Short, immediate proofs — direct and by contrapositive, paired so you see both routes to the same fact.

1. Prove directly: “If n is even, then n^2 is even.”
2. Prove directly: “If n is odd, then n^2 is odd.”
3. Prove directly: “If a and b are both even, then $a + b$ is even.”
4. State the contrapositive of “If n is a multiple of 6, then n is even.” Prove the contrapositive directly.
5. Prove directly: “If a and b are both odd, then ab is odd.”
6. True or false: “To prove ‘if P then Q ’ by contrapositive, you should assume Q is false and show P is false.”
7. Prove by contrapositive: “If n^2 is odd, then n is odd.”
8. True or false: “A proof by contrapositive is a different technique from proving the converse.”
9. Prove directly: “If $x > 1$, then $x^2 > 1$.”
10. Prove by contrapositive: “If $x^2 \leq 1$, then $x \leq 1$.”

Section B Manipulation Drills

(≤50 s each)

Slightly longer proofs, and judgement calls about which route is more natural.

1. Prove directly: “If n is a multiple of 4, then n^2 is a multiple of 16.”
2. Prove by contrapositive: “If n^2 is not a multiple of 3, then n is not a multiple of 3.”
3. Which proof route is more natural for “If n^2 is even, then n is even” — direct or contrapositive? Prove it using that route.

4. Prove directly: “If $a > b > 0$, then $a^2 > b^2$.”
5. Prove by contrapositive: “If $x^2 - 4x + 3 \neq 0$, then $x \neq 1$ and $x \neq 3$.”
6. Prove directly: “If n is a multiple of 3, then n^3 is a multiple of 27.”
7. State and prove the contrapositive of “If xy is irrational, then x is irrational or y is irrational.”
8. Prove directly: “If n ends in the digit 5, then n^2 ends in the digit 5.”
9. Prove by contrapositive: “If n is not a multiple of 4, then n^2 is not a multiple of 16.”
10. True or false: “Every proof by contrapositive can be rewritten as a direct proof of the original statement.”

Section C Substitution & Structure(≤ 90 s each)*Multi-step proofs, and your first “which condition is sufficient” MCQ.*

1. For non-zero real numbers x, y , which one of the following is sufficient to conclude that $x < y$?
 - A) $x^2 < y^2$
 - B) $y^2 < x^2$
 - C) $|x| < |y|$
 - D) $|y| < |x|$
 - E) $x^5 < y^5$
 - F) $y^5 < x^5$
2. Prove by contrapositive: “If n^3 is even, then n is even.”
3. Prove directly: “If $x, y > 0$ and $x^2 > y^2$, then $x > y$.”
4. Prove directly: “If n is a multiple of 3 but not a multiple of 9, then n^2 is a multiple of 9 but not a multiple of 81.”
5. Prove by contrapositive: “If x and y are real numbers with xy irrational, then $x \neq 0$ and $y \neq 0$.”
6. Prove by contrapositive: “If n is an integer such that $2n^2 + 1$ is a multiple of 3, then n is not a multiple of 3.”
7. Prove directly: “If n is a positive integer that is not a multiple of 3, then $n^2 \equiv 1 \pmod{3}$.”
8. True or false, with justification: “A statement with no counterexamples found after checking 100 cases has therefore been proven true by direct proof.”

Section D Challenge

(TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. A region is defined by the inequalities $x + y > 10$ and $x - y > 2$. Consider the three statements:
I. $x > 6$ II. $y > 4$ III. $(x + y)(x - y) > 20$. Which of the above statements is/are true for every point in the region?
A) none
B) I only
C) II only
D) III only
E) I and II only
F) I and III only
G) II and III only
H) I, II and III
2. Prove by contradiction: “There is no pair of positive integers (m, n) with $m^2 - n^2 = 2018$.”
3. Prove directly: “If $n > 4$ is such that both $n - 1$ and $n + 1$ are prime, then n is a multiple of 6.”
4. Prove directly: “If x and y are positive reals with $x \neq y$, then $\frac{x + y}{2} > \sqrt{xy}$.”
5. Statement T : “If n is a positive integer such that n^2 is a multiple of 6, then n is a multiple of 6.” (a) Is T true? Prove or disprove. (b) Contrast with the fact that “ n^2 is a multiple of 4” does *not* force “ n is a multiple of 4” (witnessed by $n = 6$). What property of 6, that 4 lacks, makes T true?

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths